

Table 1: Height of the spring's lower coil above the floor for different hanging masses

| Hanging mass $m$ (g) | Lower-coil height $x$ (cm) |
|----------------------|----------------------------|
| 9.76                 | 82.08                      |
| 19.84                | 74.50                      |
| 29.86                | 67.70                      |
| 39.88                | 59.05                      |
| 49.92                | 51.40                      |
| 59.94                | 43.70                      |

Table 2: Additional restoring force and extension relative to the first measurement

| Restoring force $\Delta F = (m - m_{\text{ref}})g$ (N) | Extension $\Delta x = x_{\text{ref}} - x$ (m) |
|--------------------------------------------------------|-----------------------------------------------|
| 0.0988                                                 | 0.0758                                        |
| 0.1970                                                 | 0.1438                                        |
| 0.2952                                                 | 0.2303                                        |
| 0.3936                                                 | 0.3068                                        |
| 0.4918                                                 | 0.3838                                        |

Table 3: Linear-fit result for the static method

| Quantity                             | Value                          |
|--------------------------------------|--------------------------------|
| Slope $k_{s,1}$ (N m <sup>-1</sup> ) | $1.260 \pm 0.023$              |
| $y$ -intercept (N)                   | $(7.8 \pm 5.8) \times 10^{-3}$ |

Table 4: Height of the spring's lower coil above the floor for different hanging masses

| Hanging mass $m$ (g) | Lower-coil height $x$ (cm) |
|----------------------|----------------------------|
| 40.08                | 82.60                      |
| 60.10                | 79.05                      |
| 100.16               | 71.80                      |
| 140.32               | 64.80                      |
| 180.42               | 57.80                      |
| 220.58               | 50.65                      |

Table 5: Additional restoring force and extension relative to the first measurement

| Restoring force $\Delta F = (m - m_{\text{ref}})g$ (N) | Extension $\Delta x = x_{\text{ref}} - x$ (m) |
|--------------------------------------------------------|-----------------------------------------------|
| 0.1962                                                 | 0.0355                                        |
| 0.5888                                                 | 0.1080                                        |
| 0.9824                                                 | 0.1780                                        |
| 1.3753                                                 | 0.2480                                        |
| 1.7689                                                 | 0.3195                                        |

Table 6: Linear-fit result for the static method

| Quantity                              | Value                           |
|---------------------------------------|---------------------------------|
| Slope $k_{s,2}$ ( $\text{N m}^{-1}$ ) | $5.553 \pm 0.021$               |
| $y$ -intercept (N)                    | $(-5.1 \pm 4.2) \times 10^{-3}$ |

Table 7: Oscillation period for different hanging masses ( $T_1$ : start of the 1st period,  $T_2$ : end of the 30th period)

| Hanging mass $m$ (g) | $T_1$ (s) | $T_2$ (s) | $30T = T_2 - T_1$ (s) | $T$ (s) |
|----------------------|-----------|-----------|-----------------------|---------|
| 19.82                | 25.68     | 50.95     | 25.27                 | 0.8423  |
| 29.86                | 7.57      | 38.08     | 30.51                 | 1.017   |
| 39.90                | 2.45      | 37.20     | 34.75                 | 1.158   |
| 50.00                | 2.90      | 41.44     | 38.54                 | 1.285   |
| 60.06                | 4.34      | 46.60     | 42.26                 | 1.409   |

Table 8: Linear-fit result for the dynamic method and comparison of the observed and reference effective spring mass

| Quantity                                   | Value             |
|--------------------------------------------|-------------------|
| Slope $s$ ( $\text{s}^2 \text{kg}^{-1}$ )  | $31.46 \pm 0.27$  |
| $y$ -intercept $b$ ( $\text{s}^2$ )        | $0.088 \pm 0.012$ |
| $k_{d,1} = 4\pi^2/s$ ( $\text{N m}^{-1}$ ) | $1.255 \pm 0.011$ |
| $m_{s,\text{obs}} = 3b/s$ (g)              | $8.4 \pm 1.2$     |
| $m_{s,\text{ref}}$ (g)                     | 10.40             |
| Error                                      | -19.46%           |

Table 9: Oscillation period for different hanging masses ( $T_1$ : start of the 1st period,  $T_2$ : end of the 30th period)

| Hanging mass $m$ (g) | $T_1$ (s) | $T_2$ (s) | $30T = T_2 - T_1$ (s) | $T$ (s) |
|----------------------|-----------|-----------|-----------------------|---------|
| 40.14                | 7.86      | 24.72     | 16.86                 | 0.562   |
| 80.24                | 4.21      | 27.62     | 23.41                 | 0.7803  |
| 120.34               | 10.16     | 38.46     | 28.30                 | 0.9433  |
| 160.30               | 16.14     | 48.71     | 32.57                 | 1.086   |
| 200.48               | 9.50      | 45.72     | 36.22                 | 1.207   |

Table 10: Linear-fit result for the dynamic method and comparison of the observed and reference effective spring mass

| Quantity                                   | Value               |
|--------------------------------------------|---------------------|
| Slope $s$ ( $\text{s}^2 \text{kg}^{-1}$ )  | $7.120 \pm 0.034$   |
| $y$ -intercept $b$ ( $\text{s}^2$ )        | $0.0336 \pm 0.0045$ |
| $k_{d,2} = 4\pi^2/s$ ( $\text{N m}^{-1}$ ) | $5.545 \pm 0.027$   |
| $m_{s,\text{obs}} = 3b/s$ (g)              | $14.2 \pm 2.0$      |
| $m_{s,\text{ref}}$ (g)                     | 13.16               |
| Error                                      | +7.67%              |

Table 11: Comparison of static and dynamic spring constants

| Quantity                       | Value             |
|--------------------------------|-------------------|
| $k_{s,1}$ (N m <sup>-1</sup> ) | $1.260 \pm 0.023$ |
| $k_{d,1}$ (N m <sup>-1</sup> ) | $1.255 \pm 0.011$ |
| $k_{s,2}$ (N m <sup>-1</sup> ) | $5.553 \pm 0.021$ |
| $k_{d,2}$ (N m <sup>-1</sup> ) | $5.545 \pm 0.027$ |

Table 12: Turning times and turning points of the whole motion (rail level)

| $t$ (s) | $x$ (m)  |
|---------|----------|
| 0.8209  | -0.01829 |
| 1.7424  | 0.01778  |
| 2.6974  | -0.01168 |
| 3.6189  | 0.01254  |
| 4.5571  | -0.00564 |
| 5.5288  | 0.00763  |

Table 13: Measured acceleration of the unloaded cart

| Cart rightward (cm s <sup>-2</sup> ) | Cart leftward (cm s <sup>-2</sup> ) |
|--------------------------------------|-------------------------------------|
| -2.11                                | -2.09                               |
| -2.11                                | -2.04                               |
| -2.08                                | -2.03                               |

Table 14: Period and friction-force analysis with the rail level

| Quantity                                                                        | Value                            |
|---------------------------------------------------------------------------------|----------------------------------|
| Period $T_{\text{obs}}$ (s)                                                     | $1.883 \pm 0.020$                |
| $M_{\text{blue}}$ (g)                                                           | 587.18                           |
| $m_{\text{eff}}$ (g)                                                            | 23.38                            |
| $k_{\text{eff}} = k_{d,1} + k_{s,2}$ (N m <sup>-1</sup> )                       | $6.808 \pm 0.023$                |
| $T_{\text{theo}} = 2\pi \sqrt{\frac{M + m_{\text{eff}}/3}{k_{\text{eff}}}}$ (s) | $1.8574 \pm 0.0032$              |
| Error of $T_{\text{obs}}$                                                       | +1.38%                           |
| $D$ (m)                                                                         | $(5.70 \pm 0.39) \times 10^{-3}$ |
| $f = k_{\text{eff}}D/4$ (N)                                                     | $(9.70 \pm 0.66) \times 10^{-3}$ |
| $a$ (m s <sup>-2</sup> )                                                        | $-0.02077 \pm 0.00014$           |
| $f' = M_{\text{blue}} a $ (N)                                                   | $0.012194 \pm 0.000083$          |
| Error of $f$ relative to $f'$                                                   | -20.46%                          |

Table 15: Turning times and turning points of the whole motion (rail tilted)

| $t$ (s) | $x$ (m)  |
|---------|----------|
| 0.7422  | 0.08702  |
| 1.6531  | -0.03609 |
| 2.5976  | 0.08159  |
| 3.5254  | -0.03104 |
| 4.4362  | 0.07659  |
| 5.3808  | -0.02602 |
| 6.2917  | 0.07096  |
| 7.2532  | -0.02079 |
| 8.1978  | 0.06590  |
| 9.1424  | -0.01555 |
| 10.0701 | 0.06091  |
| 11.0147 | -0.01033 |
| 11.9255 | 0.05592  |

Table 16: Period and friction-force analysis with the rail tilted

| Quantity                                    | Value                              |
|---------------------------------------------|------------------------------------|
| Period $T_{\text{obs}}$ (s)                 | $1.864 \pm 0.011$                  |
| $M'_{\text{blue}}$ (g)                      | 599.86                             |
| $T_{\text{theo}}$ (s)                       | $1.8771 \pm 0.0032$                |
| Error of $T_{\text{obs}}$                   | -0.70%                             |
| $D$ (m)                                     | $(5.168 \pm 0.063) \times 10^{-3}$ |
| $f = k_{\text{eff}}D/4$ (N)                 | $(8.80 \pm 0.11) \times 10^{-3}$   |
| $\theta$ ( $^{\circ}$ )                     | 43.7631                            |
| $f'' = M'_{\text{blue}} a  \cos \theta$ (N) | $(8.997 \pm 0.061) \times 10^{-3}$ |
| Error of $f$ relative to $f''$              | -2.22%                             |

Table 17: Curve-fitting and derived results, magnet at the higher position

| Quantity                                                 | Value                               |
|----------------------------------------------------------|-------------------------------------|
| $m_{\text{eff}}$ (g)                                     | 26.32                               |
| $k_{\text{eff}} = 2k_{s,2}$ (N m $^{-1}$ )               | $11.107 \pm 0.041$                  |
| $M_{\text{red}}$ (g)                                     | 446.64                              |
| $M_{\text{eff}} = M_{\text{red}} + m_{\text{eff}}/3$ (g) | 455.41                              |
| $A$                                                      | $0.10049 \pm 0.00025$               |
| $B$                                                      | $4.9031 \pm 0.0020$                 |
| $C$                                                      | $-2.8264 \pm 0.0023$                |
| $D$                                                      | $(-5.481 \pm 0.046) \times 10^{-3}$ |
| $E$                                                      | $-0.5969 \pm 0.0021$                |
| $\omega_0$ (rad s $^{-1}$ )                              | $4.9393 \pm 0.0020$                 |
| Damping ratio $\zeta$                                    | $0.12084 \pm 0.00042 (< 1)$         |
| $M$ (g)                                                  | $455.3 \pm 1.8$                     |
| Error of $M$ relative to $M_{\text{eff}}$                | -0.04%                              |
| Damping coefficient $c$ (kg s $^{-1}$ )                  | $0.5435 \pm 0.0028$                 |
| Quasi-period $T$ (s)                                     | $1.28146 \pm 0.00053$               |

Table 18: Curve-fitting and derived results, magnet at the lower position

| Quantity                                  | Value                               |
|-------------------------------------------|-------------------------------------|
| $A$                                       | $0.08526 \pm 0.00025$               |
| $B$                                       | $4.8264 \pm 0.0044$                 |
| $C$                                       | $-2.7911 \pm 0.0028$                |
| $D$                                       | $(-2.234 \pm 0.043) \times 10^{-3}$ |
| $E$                                       | $-1.1118 \pm 0.0043$                |
| $\omega_0$ (rad s $^{-1}$ )               | $4.9527 \pm 0.0042$                 |
| Damping ratio $\zeta$                     | $0.22448 \pm 0.00088$ ( $< 1$ )     |
| $M$ (g)                                   | $452.8 \pm 1.9$                     |
| Error of $M$ relative to $M_{\text{eff}}$ | $-0.58\%$                           |
| Damping coefficient $c$ (kg s $^{-1}$ )   | $1.0068 \pm 0.0056$                 |
| Quasi-period $T$ (s)                      | $1.3018 \pm 0.0012$                 |

Table 19: Turning times and turning points of the whole motion (equal-mass blue cart)

| $t$ (s) | $x$ (m)  |
|---------|----------|
| 0.5696  | 0.02675  |
| 1.2230  | -0.12872 |
| 1.8597  | 0.02261  |
| 2.4963  | -0.12412 |
| 3.1330  | 0.01842  |
| 3.7696  | -0.12053 |
| 4.4063  | 0.01465  |
| 5.0430  | -0.11649 |
| 5.6796  | 0.01100  |
| 6.3163  | -0.11289 |
| 6.9529  | 0.00734  |
| 7.5896  | -0.10905 |
| 8.2262  | 0.00344  |
| 8.8629  | -0.10465 |
| 9.4828  | -0.00003 |
| 10.1194 | -0.10182 |
| 10.7728 | -0.00335 |
| 11.3927 | -0.09827 |

Table 20: Period and friction-force analysis for the equal-mass blue cart

| Quantity                          | Value                              |
|-----------------------------------|------------------------------------|
| Period $T_{\text{obs}}$ (s)       | $1.2733 \pm 0.0041$                |
| $M''_{\text{blue}}$ (g)           | 447.76                             |
| $T_{\text{theo}}$ (s)             | $1.2739 \pm 0.0024$                |
| Error of $T_{\text{obs}}$         | -0.04%                             |
| $D$ (m)                           | $(3.78 \pm 0.11) \times 10^{-3}$   |
| $f = k_{\text{eff}}D/4$ (N)       | $0.01051 \pm 0.00031$              |
| $f''' = M''_{\text{blue}} a $ (N) | $(9.298 \pm 0.063) \times 10^{-3}$ |
| Error of $f$ relative to $f'''$   | +13.00%                            |

Table 21: Quasi-period of the cart in different configurations

| Configuration                    | Quasi-period (s)      |
|----------------------------------|-----------------------|
| Magnet at the lower position     | $1.3018 \pm 0.0012$   |
| Magnet at the higher position    | $1.28146 \pm 0.00053$ |
| No magnet (equal-mass blue cart) | $1.2733 \pm 0.0041$   |

Table 22: Comparison of the relative error of the friction force under different conditions

| Configuration                    | Relative error of $f$ |
|----------------------------------|-----------------------|
| Rail level                       | $-20.46\%$            |
| Rail tilted                      | $-2.22\%$             |
| Equal-mass blue cart (no magnet) | $+13.00\%$            |

Table 23: Comparison of the relative error of the oscillation period under different conditions

| Configuration                    | Relative error of $T_{\text{obs}}$ |
|----------------------------------|------------------------------------|
| Rail level                       | $+1.38\%$                          |
| Rail tilted                      | $-0.70\%$                          |
| Equal-mass blue cart (no magnet) | $-0.04\%$                          |